

PetroPro – WhatsApp Invoice Module (Claude Prompt)

Objective

Build a modular Communication Service for PetroPro without changing existing billing logic. Support both Community (Free) and Professional editions.

Community Edition (Phase 1)

After bill generation show: Print, Download PDF, Send to WhatsApp. If a registered customer is selected, use the WhatsApp/mobile number from Customer Master. If no customer is selected, prompt the operator to enter a mobile number. Generate/reuse the invoice PDF, open WhatsApp Web (wa.me) with a pre-filled thank-you message, then display a notice asking the operator to attach the generated PDF manually. Do not use browser automation or unofficial WhatsApp automation.

Professional Edition (Future)

Create a provider-based communication architecture with interfaces for WhatsApp Business API, Email, SMS and future channels. Leave WhatsApp Business provider as 'Coming Soon' but fully architected.

Architecture

Billing -> CommunicationService -> Provider. Create ICommunicationProvider, WhatsAppCommunityProvider, WhatsAppBusinessProvider, EmailProvider, SMSProvider, CommunicationLog model and CommunicationResult model.

Customer Experience

Support: 1. Walk-in customers: enter mobile number. 2. Registered customers: auto-fill from Customer Master. 3. Allow operator to edit the number before sending.

Configuration

Add Settings > Communication with Enable WhatsApp, Default Country Code, Message Template, Auto Open WhatsApp and Future Business API toggle.

Communication Log

Store InvoiceId, CustomerId, Mobile Number, Channel, Status, Date/Time and Remarks.

Validation

Validate mobile number, PDF generation, popup blocking and missing customer details. Show friendly messages.

UI

Add a generic 'Send Invoice' button with channels: WhatsApp (active), Email (future), SMS (future), Share PDF.

Development Requirements

Use clean architecture, SOLID principles, TypeScript interfaces, reusable components, responsive UI, tests, documentation and environment variables for future API keys.